MEMORY SYSTEM

Basic Concepts:

Address	Memory Locations
16 Bit	2^{16} = 64 K
32 Bit	2^{32} = 4G (Giga)
40 Bit	2^{40} = IT (Tera)

- Maximum size of memory that can be used in any computer is determined by addressing mode.
- If MAR is k-bits long then

  → memory may contain upto 2^K addressable-locations
- If MDR is n-bits long, then

  → n-bits of data are transferred between the memory and processor.
- The data-transfer takes place over the processor-bus (Figure).

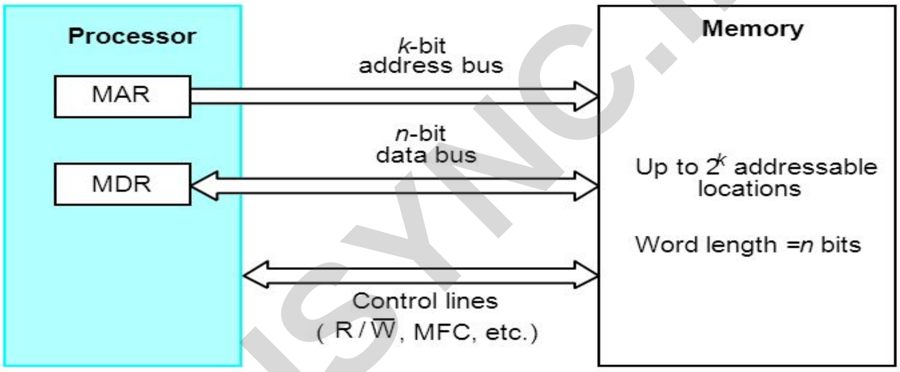


Figure Connection of the memory to the processor.

- The processor-bus has
  1) Address-Line
  2) Data-line &
  3) Control-Line (R/W, MFC – Memory Function Completed).
- The Control-Line is used for coordinating data-transfer.
- The processor reads the data from the memory by

  → loading the address of the required memory-location into MAR and

  → setting the R/W line to 1.
- The memory responds by

  → placing the data from the addressed-location onto the data-lines and

  → confirms this action by asserting MFC signal.
- Upon receipt of MFC signal, the processor loads the data from the data-lines into MDR.
- The processor writes the data into the memory-location by

→ loading the address of this location into MAR &

→ setting the R/W line to 0.

- **Memory Access Time:** It is the time that elapses between
  → Initiation of an operation & Completion of that operation.

- **Memory Cycle Time:** It is the minimum time delay that required between the initiation of the two successive memory-operations.

RAM (Random Access Memory)

- In RAM, any location can be accessed for a Read/Write-operation in fixed amount of time,

Cache Memory

- It is a small, fast memory that is inserted between larger slower main-memory and processor.
- It holds the currently active segments of a program and their data.

Virtual Memory

- The address generated by the processor is referred to as a **virtual/logical address**.
- The virtual-address-space is mapped onto the physical-memory where data are actuallystored.
- The mapping-function is implemented by MMU (MMU = memory management unit).
- Only the active portion of the address-space is mapped into locations in the physical-memory.
- The remaining virtual-addresses are mapped onto the bulk storage devices such as magneticdisk.
- As the active portion of the virtual-address-space changes during program execution, the MMU changes the mapping-function & transfers the data between disk and memory.
- During every memory-cycle, MMU determines whether the addressed-page is in the memory.
- If the page is in the memory. Then, the proper word is accessed and execution proceeds. Otherwise, a page containing desired word is transferred from disk to memory.
- .

- Memory can be classified as follows:

1) RAM which can be further classified as follows:
   i) Static RAM
   ii) Dynamic RAM (DRAM)
2) ROM which can be further classified as follows:
   i) PROM
   ii) EPROM
   iii) EEPROM &
   iv) Flash Memory which can be further classified as Flash Cards & Flash Drives.

Semi-Conductor RAM Memories:

Internal Organization of Memory-Chips:

- Memory-cells are organized in the form of array.
- Each cell is capable of storing 1-bit of information.
- Each row of cells forms a memory-word.
- All cells of a row are connected to a common line called as **Word-Line**.
- The cells in each column are connected to **Sense/Write** circuit by 2-bit-lines.
- The Sense/Write circuits are connected to data-input or output lines of the chip.

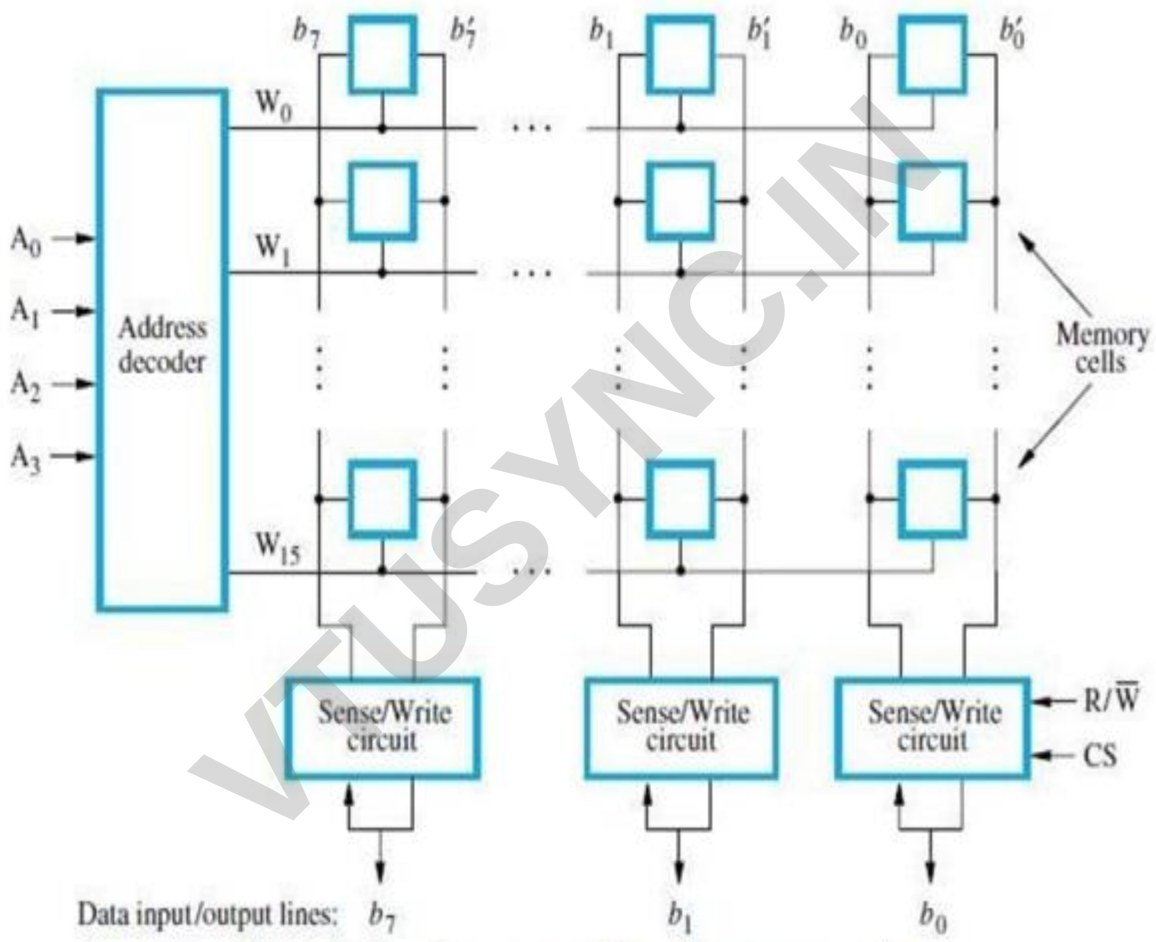


Figure Organization of bit cells in a memory chip.

- During a write-operation, the sense/write circuit

  → receive input information &

  → Store input info in the cells of the selected word.
- The data-input and data-output of each Sense/Write circuit are connected to a single bidirectionaldata-line.
- Data-line can be connected to a data-bus of the computer.
- Following 2 control lines are also used:

  1) **R/W’** specifies the required operation.

Bit Organization	Requirement of external connection for address, data and control lines
128 (16x8)	14
(1024) 128x8(1k)	19

2) **CS’** □ Chip Select input selects a given chip in the multi-chip memory-system.

<u>Static Memories:</u>

STATIC RAM:

- Memories consist of circuits capable of retaining their state as long as power is applied are known**.**

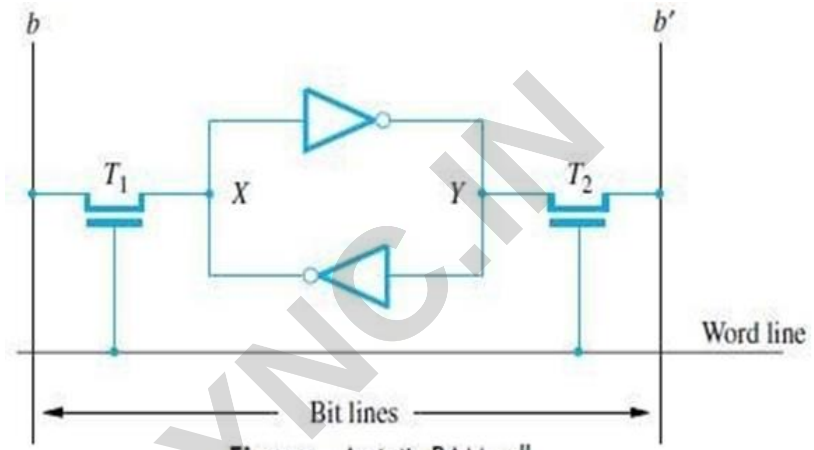


Figure A static RAM cell.

- Two inverters are cross connected to form a latch (Figure).
- The latch is connected to 2-bit-lines by transistors T1 and T2.
- The transistors act as switches that can be opened/closed under the control of the word-line.
- When the word-line is at ground level, the transistors are turned off and the latch retain its state.

Read Operation

- To read the state of the cell, the word-line is activated to close switches T1 and T2.
- If the cell is in state 1, the signal on bit-line b is high and the signal on the bit-line b’ is low.
- Thus, b and b’ are complement of each other.
- Sense/Write circuit

  → monitors the state of b & b’ and sets the output accordingly.

Write Operation

- The state of the cell is set by

  → placing the appropriate value on bit-line b and its complement on b’ and

  → then activating the word-line. This forces the cell into the corresponding state.
- The required signal on the bit-lines is generated by Sense/Write circuit.

CMOS Cell:

- Transistor pairs (T3, T5) and (T4, T6) form the inverters in the latch (Figure).
- In state 1, the voltage at point X is high by having T5, T4 ON and T3, T6 are OFF.
- Thus, T1 and T2 returned ON (Closed), bit-line b and b' will have high and low signals respectively.

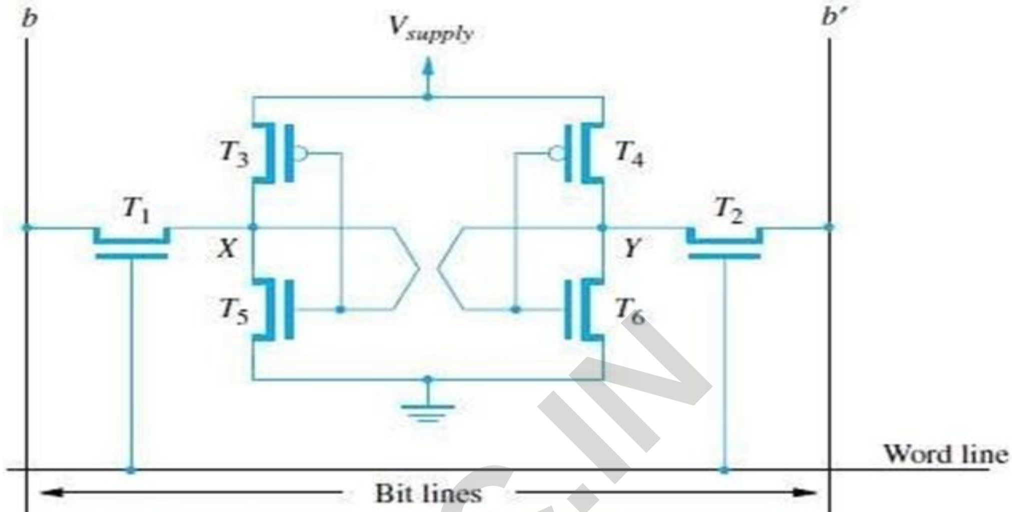


Figure An example of a CMOS memory cell.

Advantages:

1. It has low power consumption " the current flows in the cell only when the cell is active".
2. Static RAM's can be accessed quickly. It access time is few nanoseconds.

Disadvantage:

- SRAMs are said to be volatile memories." their contents are lost when poweris interrupted".

Asynchronous DRAM:

- Less expensive RAMs can be implemented if simple cells are used.
- Such cells cannot retain their state indefinitely. Hence they arecalled **Dynamic RAM (DRAM).**

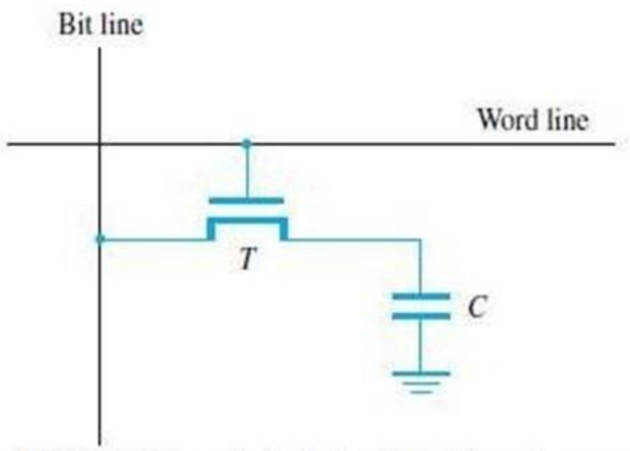


Figure A single-transistor dynamic memory cell.

- The information stored in a dynamic memory-cell in the form of a charge on a capacitor.
- This charge can be maintained only for tens of milliseconds.
- The contents must be periodically refreshed by restoring this capacitor charge to its full value.
- In order to store information in the cell, the transistor T is turned “ON‟ (Figure).
- The appropriate voltage is applied to the bit-line which charges the capacitor.
- After the transistor is turned off, the capacitor begins to discharge.
- Hence, information Stored in cell can be retrieved correctly before threshold value of capacitor drops down.
- During a read-operation,

  → Transistor is turned “ON‟

  → a sense amplifier detects whether the charge on the capacitor is above the threshold value.

➢ If (charge on capacitor) > (threshold value) □ Bit-line will have logic value “1‟.

➢ If (charge on capacitor) < (threshold value) □ Bit-line will set to logic value “0‟.

Asynchronous DRAM Description

- The 4 bit cells in each row are divided into 512 groups of 8 (Figure).

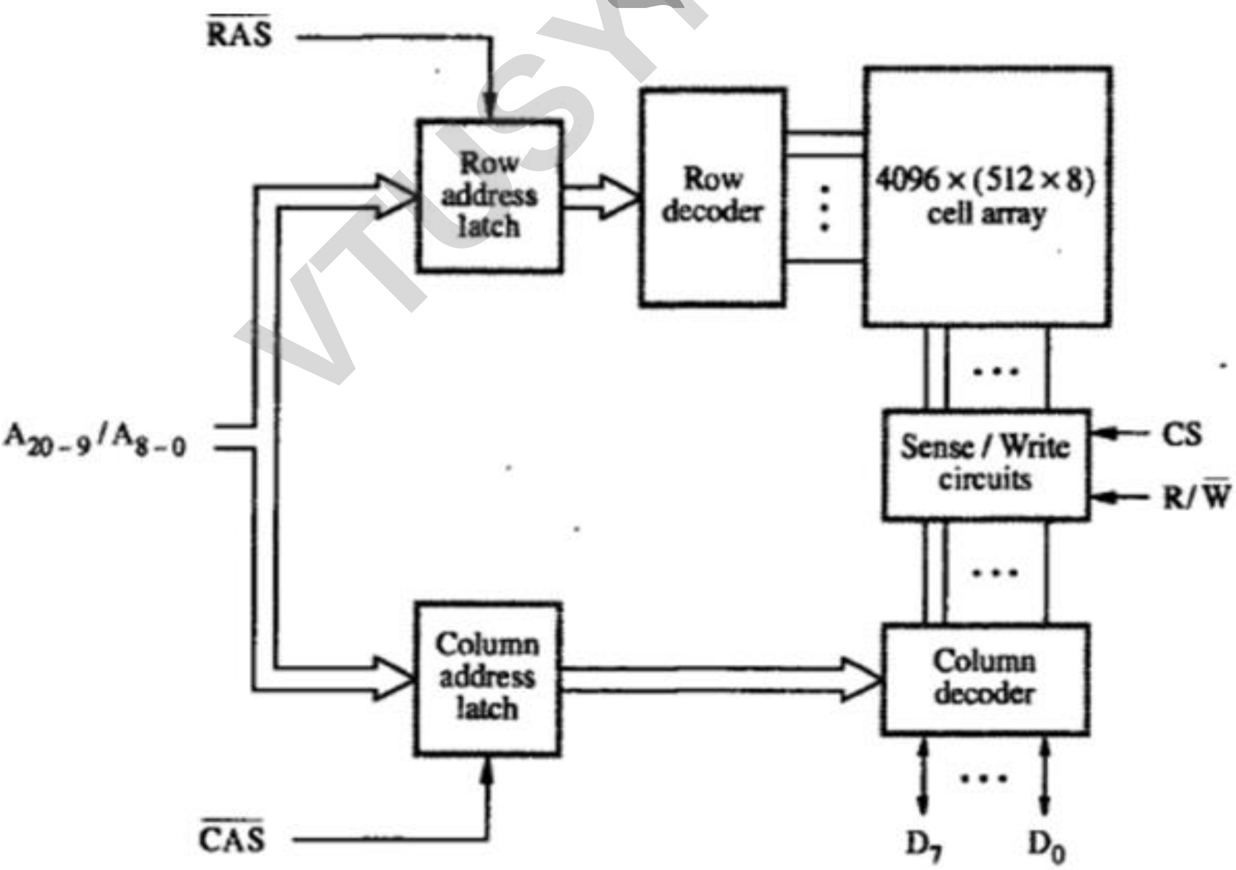


Figure Internal organization of a 2M × 8 dynamic memory chip

- 21 bit address is needed to access a byte in the memory.
- 21 bit is divided as follows:

  12 address bits are needed to select a row.

  i.e. A8-0 → specifies row-address of a byte.

  9 bits are needed to specify a group of 8 bits in the selected row.

  i.e. A20-9 → specifies column-address of a byte.

- During Read/Write-operation,

  → row-address is applied first.

  → row-address is loaded into row-latch in response to a signal pulse on **RAS’** input of chip.(RAS = Row-address Strobe CAS = Column-address Strobe)

- When a Read-operation is initiated, all cells on the selected row are read and refreshed.
- Shortly after the row-address is loaded, the column-address is applied to the address pins & loaded into **CAS’.**
- The information in the latch is decoded. The appropriate group of 8 Sense/Write circuits is selected.

  R/W’=1(read-operation) □ Output values of selected circuits are transferred to data-lines D0-D7.

  R/W’=0(write-operation) □ Information on D0-D7 are transferred to the selected circuits.

- RAS‟ & CAS‟ are active-low so that they cause latching of address when they change from high to low.
- To ensure that the contents of DRAMs are maintained, each row of cells is accessedperiodically.
- A special memory-circuit provides the necessary control signals RAS & CAS that govern the timing.
- The processor must take into account the delay in the response of the memory.

READ ONLY MEMORY (ROM)

- Both SRAM and DRAM chips are volatile, i.e. they lose the stored information if power is turned off.
- Many applications require non-volatile memory which retains the stored information if power isturned off.
- For ex:

  OS software has to be loaded from disk to memory i.e. it requires non-volatile memory.

- Non-volatile memory is used in embedded system.
- Since the normal operation involves only reading of stored data, a memory of this type is called ROM.
  - **At Logic value ‘0’** □ Transistor (T) is connected to the ground point(P).

    Transistor switch is closed & voltage on bit-line nearly drops to zero (Figure).

  - **At Logic value ‘1’** □ Transistor switch is open.

    The bit-line remains at high voltage.

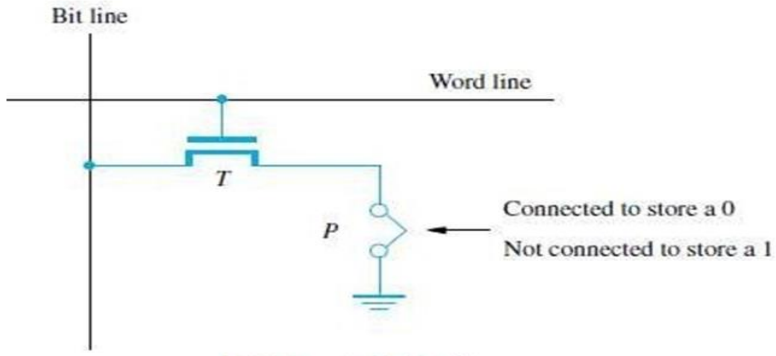


Figure A ROM cell.

- To read the state of the cell, the word-line is activated.
- A Sense circuit at the end of the bit-line generates the proper output value.

TYPES OF ROM:

- Different types of non-volatile memory are
  1. PROM
  2. EPROM
  3. EEPROM &
  4. Flash Memory (Flash Cards & Flash Drives)

PROM (PROGRAMMABLE ROM)

- PROM allows the data to be loaded by the user.
- Programmability is achieved by inserting a "fuse" at point P in a ROM cell.
- Before PROM is programmed, the memory contains all 0's.
- User can insert 1's at required location by burning-out fuse using high current-pulse.
- This process is irreversible.
- **Advantages:**
  1) It provides flexibility.
  2) It is faster.
  3) It is less expensive because they can be programmed directly by the user.

EPROM (ERASABLE REPROGRAMMABLE ROM)

- EPROM allows
  → stored data to be erased and
  → new data to be loaded.
- In cell, a connection to ground is always made at P and a special transistor is used.

- The transistor has the ability to function as
  → a normal transistor or
  → a disabled transistor that is always turned off.
- Transistor can be programmed to behave as a permanently open switch, by injecting charge into it.
- Erasure requires dissipating the charges trapped in the transistor of memory-cells. This can be done by exposing the chip to ultra-violet light.
- **Advantages:**
  1) It provides flexibility during the development-phase of digital-system.
  2) It is capable of retaining the stored information for a long time.
- **Disadvantages:**
  1) The chip must be physically removed from the circuit for reprogramming.
  2) The entire contents need to be erased by UV light.

EEPROM (ELECTRICALLY ERASABLE ROM)

- **Advantages:**
  1) It can be both programmed and erased electrically.
  2) It allows the erasing of all cell contents selectively.
- **Disadvantage:** It requires different voltage for erasing, writing and reading the stored data.

FLASH MEMORY

- In EEPROM, it is possible to read & write the contents of a single cell.
- In Flash device, it is possible to read contents of a single cell & write entire contents of a block.
- Prior to writing, the previous contents of the block are erased.
  Eg. In MP3 player, the flash memory stores the data that represents sound.
- Single flash chips cannot provide sufficient storage capacity for embedded-system.
- **Advantages:**
  1) Flash drives have greater density which leads to higher capacity & low cost per bit.
  2) It requires single power supply voltage & consumes less power.
- There are 2 methods for implementing larger memory: 1) Flash Cards & 2) Flash Drives

  1) Flash Cards
  - One way of constructing larger module is to mount flash-chips on a small card.
  - Such flash-card have standard interface.
  - The card is simply plugged into a conveniently accessible slot.
  - Memory-size of the card can be 8, 32 or 64MB.

- Eg: A minute of music can be stored in 1MB of memory. Hence 64MB flash cards can store an hour of music.

2) Flash Drives

- Larger flash memory can be developed by replacing the hard disk-drive.
- The flash drives are designed to fully emulate the hard disk.
- The flash drives are solid state electronic devices that have no movable parts.

Advantages:

1) They have shorter seek & access time which results in faster response.
2) They have low power consumption.
3) They are insensitive to vibration.

Disadvantages:

1) The capacity of flash drive (<1GB) is less than hard disk (>1GB).
2) It leads to higher cost per bit.

CACHE MEMORIES

- The effectiveness of cache mechanism is based on the property of **Locality of Reference.**

Locality of Reference

- Many instructions in the localized areas of program are executed repeatedly during some time period
- Remainder of the program is accessed relatively infrequently.
- There are 2 types:

1) Temporal

- The recently executed instructions are likely to be executed again very soon.

2) Spatial

- Instructions in close proximity to recently executed instruction are also likely to be executed soon.

- If active segment of program is placed in cache-memory, then total execution time can be reduced.
- **Block** refers to the set of contiguous address locations of some size.
- The cache-line is used to refer to the cache-block.

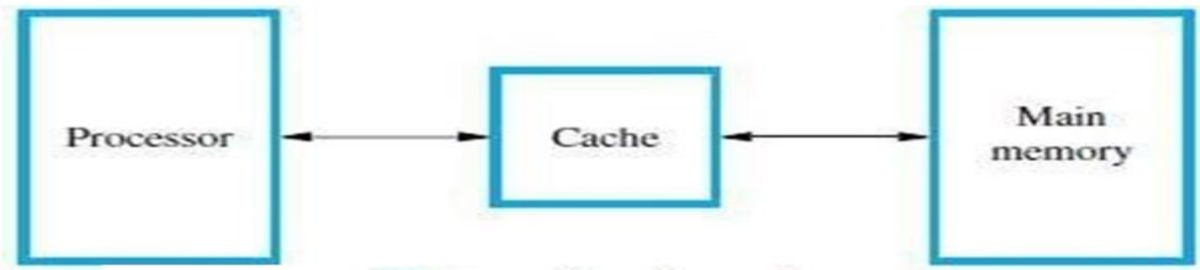


Figure Use of a cache memory.

- The Cache-memory stores a reasonable number of blocks at a given time.
- This number of blocks is small compared to the total number of blocks available in main-memory.
- Correspondence b/w main-memory-block & cache-memory-block is specified by mapping-function.
- Cache control hardware decides which block should be removed to create space for the new block.
- The collection of rule for making this decision is called the **Replacement Algorithm.**
- The cache control-circuit determines whether the requested-word currently exists in the cache.
- The write-operation is done in 2 ways: 1) Write-through protocol & 2) Write-back protocol.

Write-Through Protocol

➢ Here the cache-location and the main-memory-locations are updated simultaneously.

Write-Back Protocol

➢ This technique is to

→ Update only the cache-location &

→ Mark the cache-location with associated flag bit called **Dirty/Modified Bit.**

➢ The word in memory will be updated later, when the marked-block is removed from cache.

During Read-operation

- If the requested-word currently not exists in the cache, then **read-miss** will occur.
- To overcome the read miss, Load–through/Early restart protocol is used.

Load–Through Protocol

➢ The block of words that contains the requested-word is copied from the memory into cache.

➢ After entire block is loaded into cache, the requested-word is forwarded to processor.

During Write-operation

- If the requested-word not exists in the cache, then **write-miss** will occur.

3) If **Write Through Protocol** is used, the information is written directly into main-memory.

4) If **Write Back Protocol** is used,

→ then block containing the addressed word is first brought into the cache &

→ then the desired word in the cache is over-written with the new information.

VIRTUAL MEMORY

- It refers to a technique that automatically move program/data blocks into the main-memory when they are required for execution (Figure 8.24).
- The address generated by the processor is referred to as a **virtual/logical address**.
- The virtual-address is translated into physical-address by **MMU** (Memory Management Unit).
- During every memory-cycle, MMU determines whether the addressed-word is in the memory.

  If the word is in memory.

    Then, the word is accessed and execution proceeds.

    Otherwise, a page containing desired word is transferred from disk to memory.
- Using DMA scheme, transfer of data between disk and memory is performed.

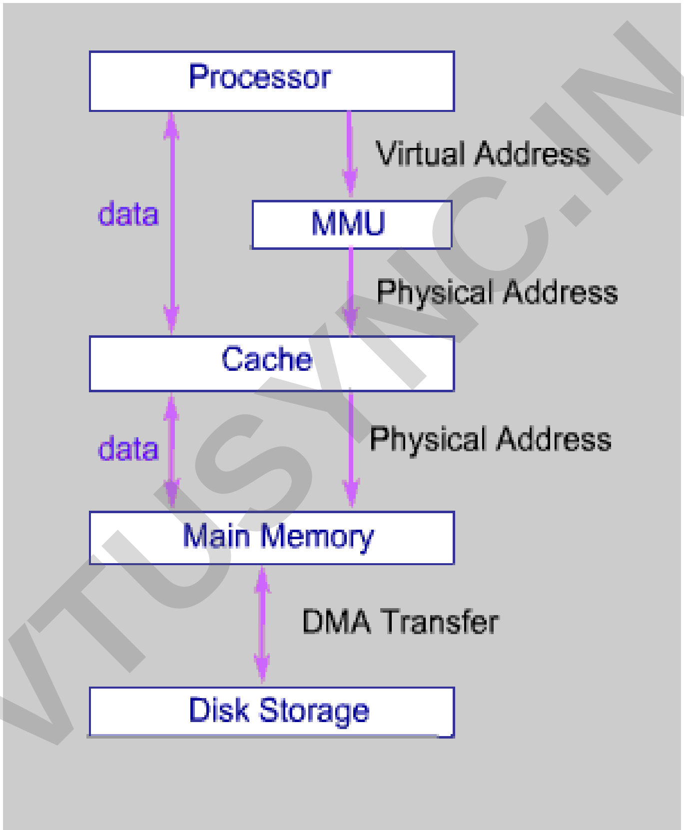


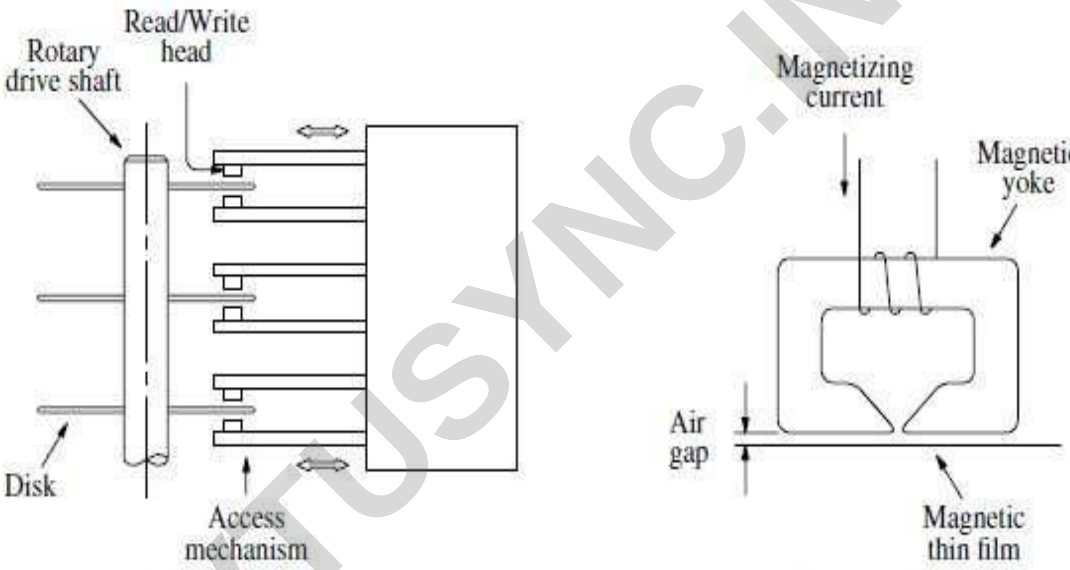
Fig: Virtual memory organization

Secondary-Storage:

- The semi-conductor memories do not provide all the storage capability.
- The secondary-storage devices provide larger storage requirements.
- Some of the secondary-storage devices are:
  1) Magnetic Disk
  2) Optical Disk &
  3) Magnetic Tapes.

Magnetic DISK

- Magnetic Disk system consists of one or more **disk** mounted on a common **spindle**.
- A **thin magnetic film** is deposited on each disk.
- Disk is placed in a **rotary-drive** so that magnetized surfaces move in close proximity to R/W heads.
- Each **R/W head** consists of 1) Magnetic Yoke & 2) Magnetizing-Coil.
- Digital information is stored on magnetic film by applying current pulse to the magnetizing-coil.
- Only changes in the magnetic field under the head can be sensed during the Read-operation.
- Therefore, if the binary states 0 & 1 are represented by two opposite states, then a voltage is induced in the head only at 0-1 and at 1-0 transition in the bit stream.
- A consecutive of 0's & 1's are determined by using the clock.
- **Manchester Encoding** technique is used to combine the clocking information with data.

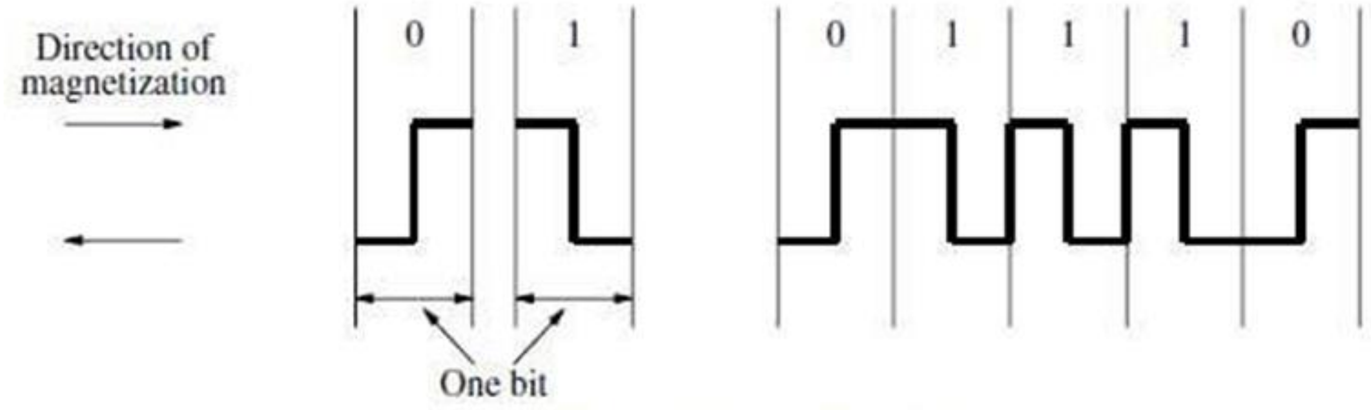


(a) Mechanical structure (b) Read/Write head detail

- R/W heads are maintained at small distance from disk-surfaces in order to achieve high bit densities.
- When disk is moving at their steady state, the air pressure develops b/w disk-surfaces & head. This air pressure forces the head away from the surface.
- The flexible spring connection between head and its arm mounting permits the head to fly at the desired distance away from the surface.

Winchester Technology

- Read/Write heads are placed in a sealed, air–filtered enclosure called the Winchester Technology.
- The read/write heads can operate closure to magnetic track surfaces because the dust particles which are a problem in unsealed assemblies are absent.



(c) Bit representation by phase encoding

Figure Magnetic disk principles.

Advantages

- It has a larger capacity for a given physical size.
- The data intensity is high because the storage medium is not exposed to contaminating elements.
- The read/write heads of a disk system are movable.
- The disk system has 3 parts: 1) Disk Platter (Usually called Disk)
  2) Disk-drive (spins the disk & moves Read/write heads)
  3) Disk Controller (controls the operation of the system)

Organization & Accessing of Data on a DISK

- Each surface is divided into concentric **Tracks** (Figure).
- Each track is divided into **Sectors.**
- The set of corresponding tracks on all surfaces of a stack of disk form a **Logical Cylinder**.
- The data are accessed by specifying the surface number, track number and the sector number.
- The Read/Write-operation starts at sector boundaries.
- Data bits are stored serially on each track.
- Each sector usually contains 512 bytes.
- **Sector Header** --> contains identification information.
  It helps to find the desired sector on the selected track.
- **ECC** (Error checking code)- is used to detect and correct errors.
- An unformatted disk has no information on its tracks.
- The formatting process divides the disk physically into tracks and sectors.

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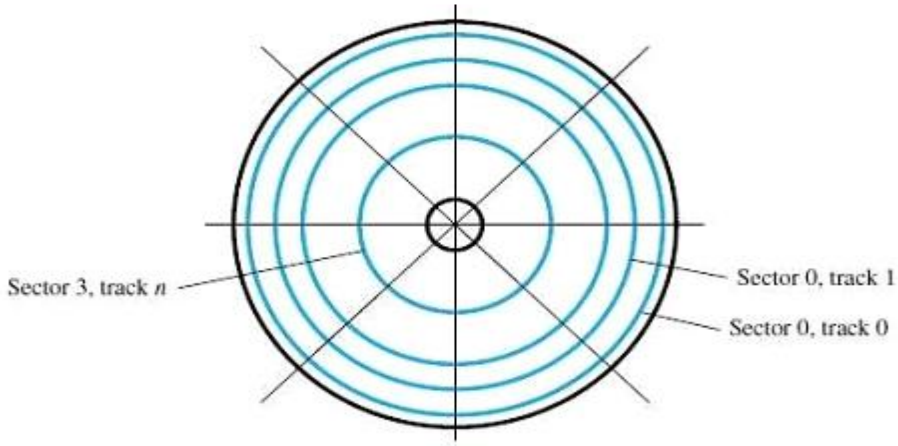

Organization of one surface of a disk.

- The formatting process may discover some defective sectors on all tracks.
- **Disk Controller** keeps a record of various defects.

DISK Controller

- The disk controller acts as interface between disk-drive and system-bus.
- The disk controller uses DMA scheme to transfer data between disk and memory.
- When the OS initiates the transfer by issuing R/W" request, the controllers register will load the following information:

  1) Memory Address: Address of first memory-location of the block of words involved in the transfer.

  2) Disk Address: Location of the sector containing the beginning of the desired block of words.

  3) Word Count: Number of words in the block to be transferred.

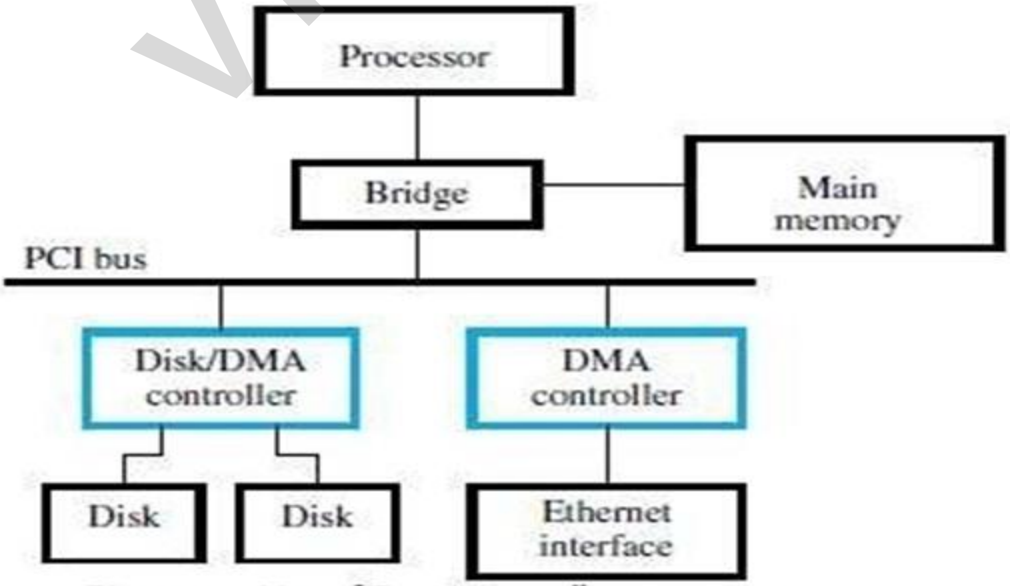

Figure Use of DMA controllers in a computer system.

- The disk-address issued by the OS is a logical address.
- The corresponding physical-address on the disk may be different.
- The controller's major functions are:

  1) Seek - Causes disk-drive to move the R/W head from its current position to desired track.

  2) Read - Initiates a Read-operation, starting at address specified in the disk-address register. Data read serially from the disk are assembled into words and placed into the data buffer for transfer to the main-memory.

  3) Write - Transfers data to the disk.

  4) Error Checking - Computes the error correcting code (ECC) value for the data read from a given sector and compares it with the corresponding ECC value read from the disk.

  In case of a mismatch, it corrects the error if possible;

  Otherwise, it raises an interrupt to inform the OS that an error has occurred.